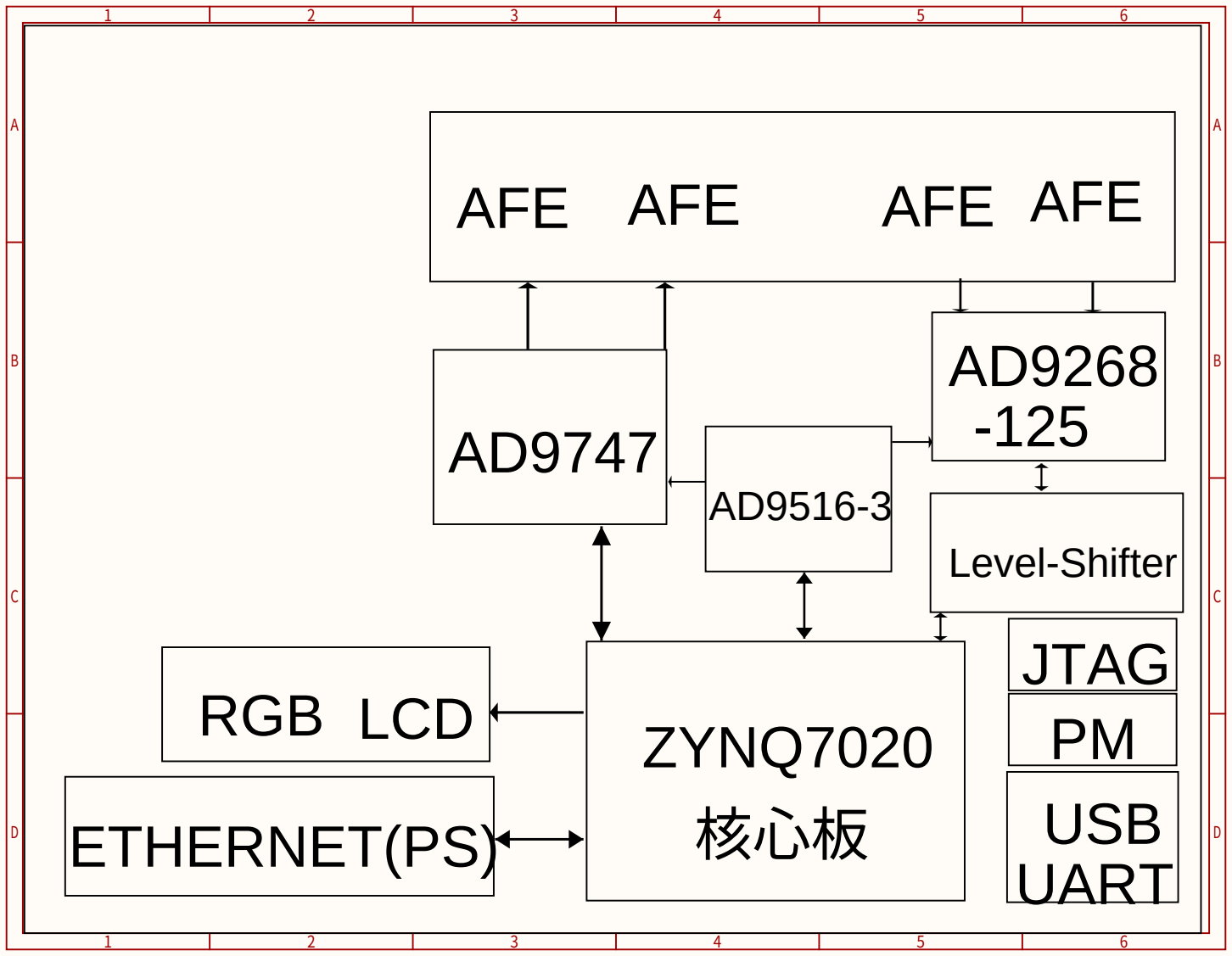
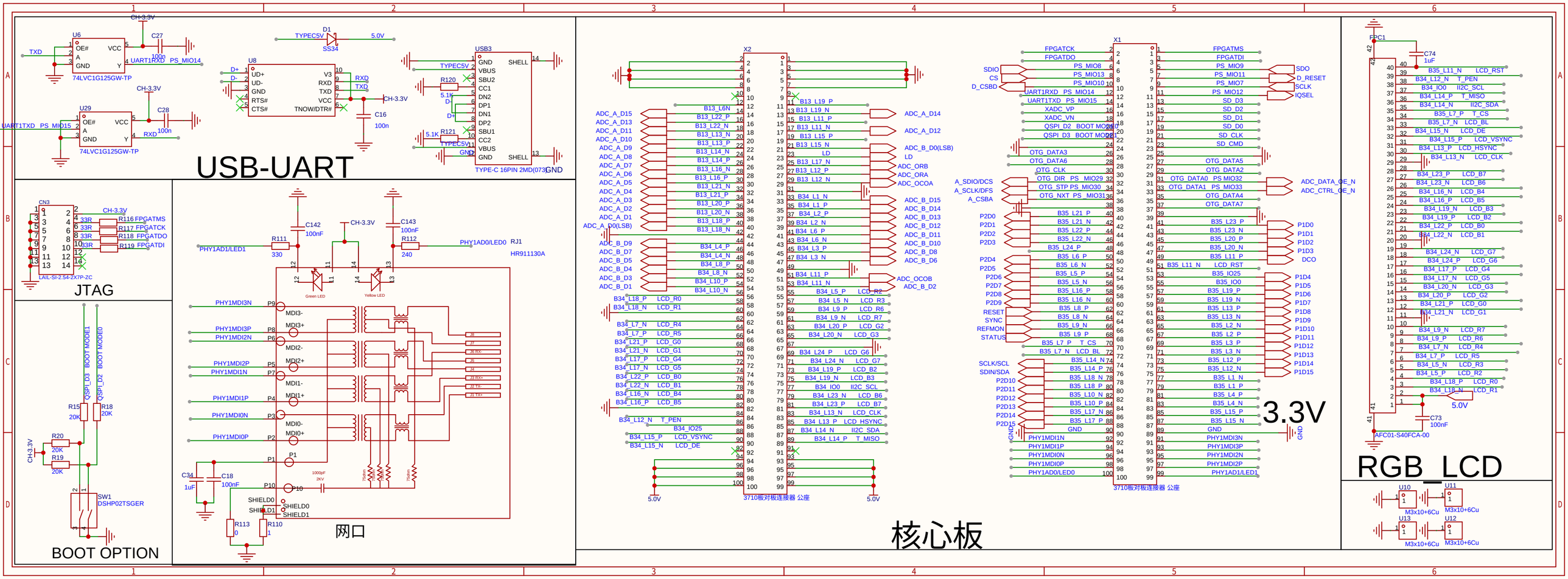
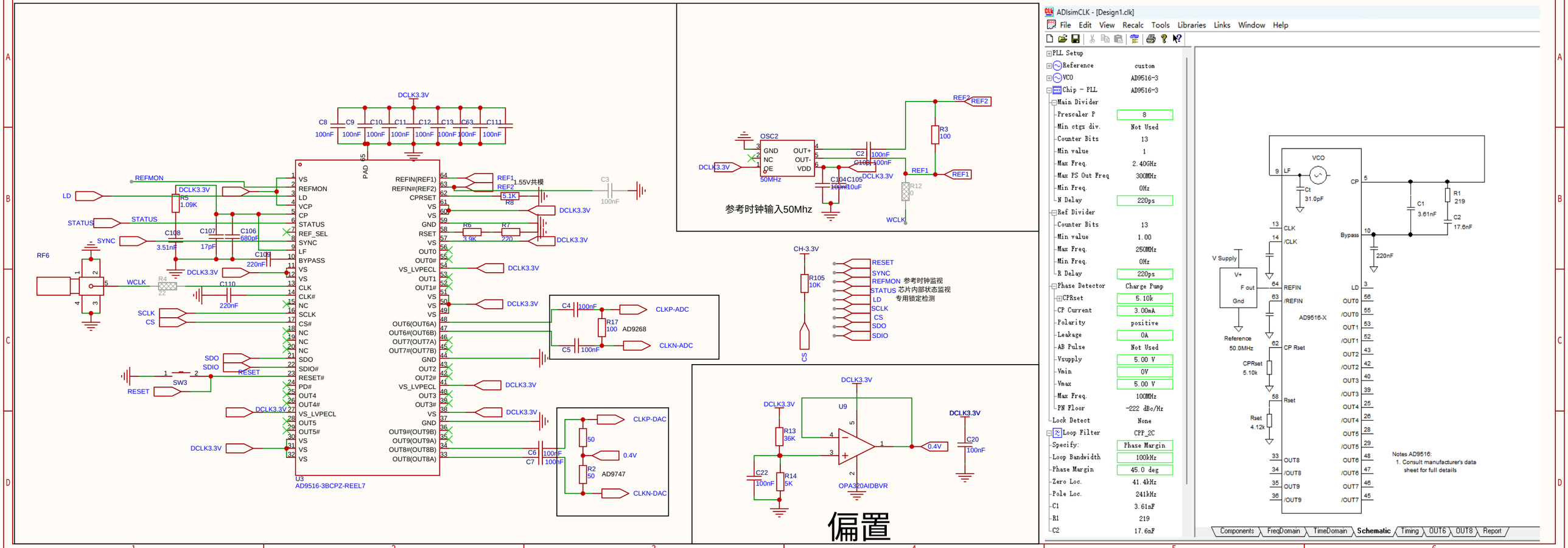

AFE AFE AFE AFE
AD9747
AD9268
-125
AD9516-3
Level-Shifter
JTAG
PM
USB
UART
RGB LCD
ETHERNET(PS)
ZYNQ7020
核心板


USB-UART
JTAG
BOOT OPTION
网口
核心板
RGB_LCD
3.3V
3710板对板连接器 公座
3710板对板连接器 公座
M3x10+6Cu
M3x10+6Cu
M3x10+6Cu
M3x10+6Cu

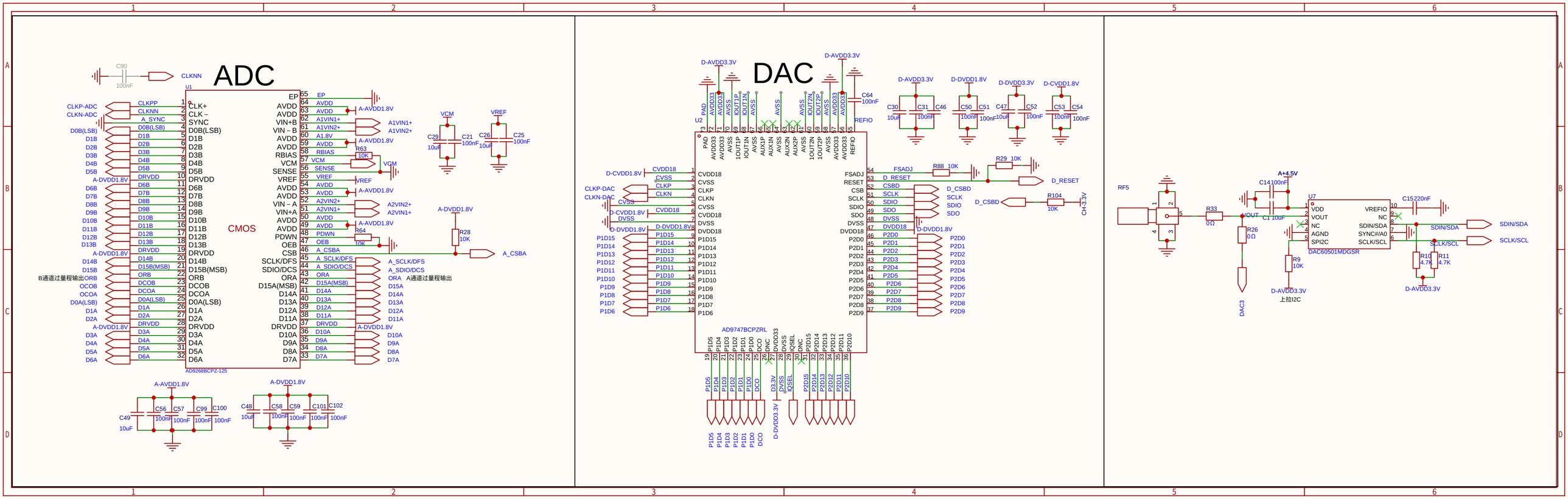


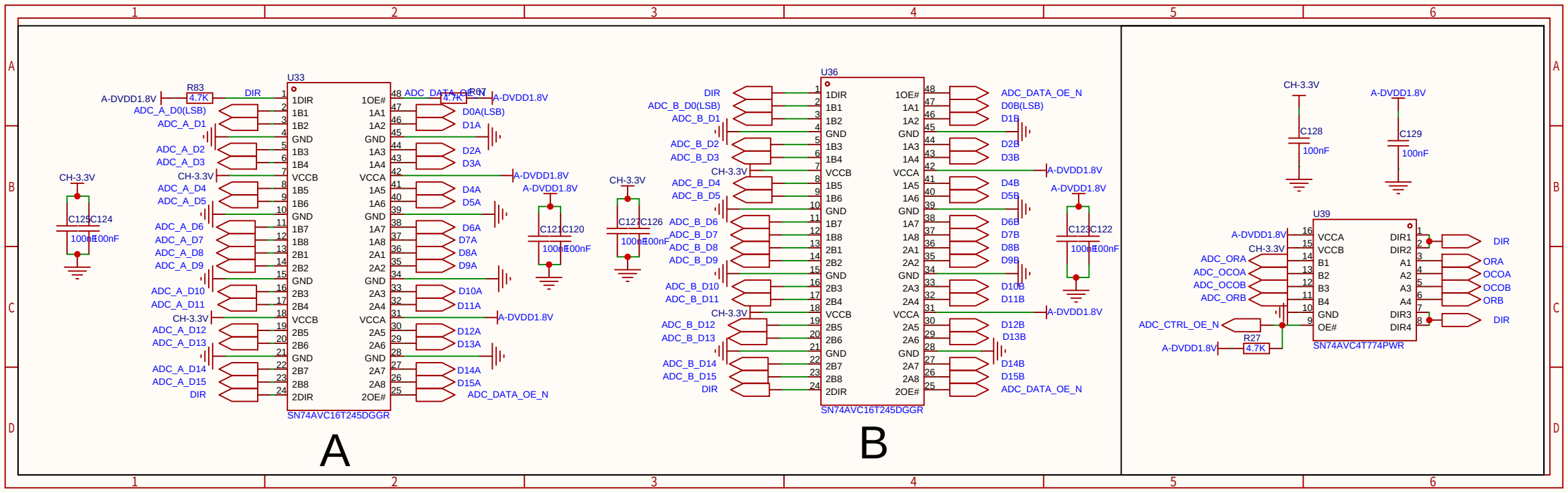
ADIsimCLK - [Design1.clk]

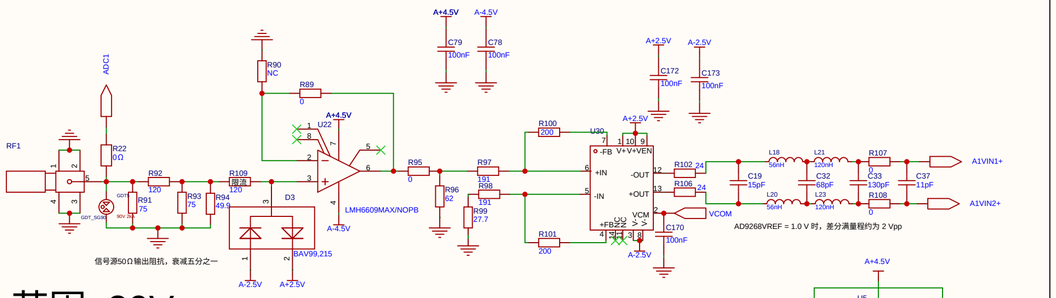
File Edit View Recalc Tools Libraries Links Window Help

PLL Setup	
Reference	custom
VCO	AD9516-3
Chip - PLL	AD9516-3
Main Divider	
Prescaler P	8
Min ctgs div.	Not Used
Counter Bits	13
Min value	1
Max Freq.	2.40GHz
Max PS Out Freq	300MHz
Min Freq.	0Hz
N Delay	220ps
Ref Divider	
Counter Bits	13
Min value	1.00
Max Freq.	250MHz
Min Freq.	0Hz
R Delay	220ps
Phase Detector	Charge Pump
CPRset	5.10k
CP Current	3.00mA
Polarity	positive
Leakage	0A
AB Pulse	Not Used
Vsupply	5.00 V
Vmin	0V
Vmax	5.00 V
Max Freq.	100MHz
PN Floor	-222 dBc/Hz
Lock Detect	None
Loop Filter	CPF_2C
Specify:	Phase Margin
Loop Bandwidth	100kHz
Phase Margin	45.0 deg
Zero Loc.	41.4kHz
Pole Loc.	241kHz
C1	3.61nF
R1	219
C2	17.6nF

Components FreqDomain TimeDomain Schematic Timing OUT6 OUT8 Report


ADC
C90
100nF
CLKNN
U1
CLKP-ADC
CLKN-ADC
D0B(LSB)
D1B
D2B
D3B
D4B
D5B
A-DVDD1.8V
D6B
D7B
D8B
D9B
D10B
D11B
D12B
D13B
D14B
D15B
B通道过量程输出ORB
OCOB
D0A(LSB)
D1A
D2A
D3A
D4A
D5A
D6A
CLKP
CLKN
A_SYNC
D0B(LSB)
D1B
D2B
D3B
D4B
D5B
DRVDD
D6B
D7B
D8B
D9B
D10B
D11B
D12B
D13B
DRVDD
D14B
D15B(MSB)
ORB
DCOB
DCOA
D0A(LSB)
D1A
D2A
DRVDD
D3A
D4A
D5A
D6A
CLK+
CLK－
SYNC
D0B(LSB)
D1B
D2B
D3B
D4B
D5B
DRVDD
D6B
D7B
D8B
D9B
D10B
D11B
D12B
D13B
DRVDD
D14B
D15B(MSB)
ORB
DCOB
DCOA
D0A(LSB)
D1A
D2A
DRVDD
D3A
D4A
D5A
D6A
CMOS
AD9268BCPZ-125
EP
AVDD
AVDD
VIN+B
VIN－B
AVDD
AVDD
RBIAS
VCM
SENSE
VREF
AVDD
AVDD
VIN－A
VIN+A
AVDD
AVDD
PDWN
OEB
CSB
SCLK/DFS
SDIO/DCS
ORA
D15A(MSB)
D14A
D13A
D12A
D11A
DRVDD
D10A
D9A
D8A
D7A
EP
AVDD
AVDD
A1VIN1
A1VIN2
A1.8V
AVDD
RBIAS
VCM
SENSE
VREF
AVDD
AVDD
A2VIN2
A2VIN1
AVDD
AVDD
PDWN
OEB
A_CSBA
A_SCLK/DFS
A_SDIO/DCS
ORA
D15A(MSB)
D14A
D13A
D12A
D11A
DRVDD
D10A
D9A
D8A
D7A
A-AVDD1.8V
A1VIN1+
A1VIN2+
A-AVDD1.8V
R63
10K
VCM
VREF
A-AVDD1.8V
A2VIN2+
A2VIN1+
A-AVDD1.8V
R64
10K
A-DVDD1.8V
R28
10K
A_CSBA
A_SCLK/DFS
A_SDIO/DCS
ORA
A通道过量程输出
D15A
D14A
D13A
D12A
D11A
D10A
D9A
D8A
D7A
VCM
C29
10uF
C21
100nF
VREF
C26
10uF
C25
100nF
A-AVDD1.8V
C49
10uF
C56
100nF
C57
100nF
C99
100nF
C100
100nF
A-AVDD1.8V
C48
10uF
C58
100nF
C59
100nF
C101
100nF
C102
100nF
DAC
D-AVDD3.3V
U2
PAD
AVDD33
AVDD33
AVSS
IOUT1P
IOUT1N
AVSS
AVSS
AUX1P
AUX1N
AVSS
AUX2N
AUX2P
AVSS
IOUT2N
IOUT2P
AVSS
AVDD33
AVDD33
REFIO
D-AVDD3.3V
C64
100nF
REFIO
D-AVDD3.3V
C30
10uF
C31
100nF
C46
D-DVDD1.8V
C50
100nF
C51
100nF
D-DVDD3.3V
C47
10uF
C52
100nF
D-CVDD1.8V
C53
100nF
C54
100nF
CVDD18
CVSS
CLKP
CLKN
CVSS
CVDD18
DVSS
D-DVDD1.8V
P1D15
P1D14
P1D13
P1D12
P1D11
P1D10
P1D9
P1D8
P1D7
P1D6
CLKP-DAC
CLKN-DAC
CVDD18
CVSS
CLKP
CLKN
CVSS
CVDD18
DVSS
D-DVDD1.8V
DVDD18
P2D0
P2D1
P2D2
P2D3
P2D4
P2D5
P2D6
P2D7
P2D8
P2D9
FSADJ
D_RESET
CSB
SCLK
SDIO
SDO
DVSS
DVDD18
P2D0
P2D1
P2D2
P2D3
P2D4
P2D5
P2D6
P2D7
P2D8
P2D9
R88 10K
R29 10K
D_RESET
D_CSBD
R104
10K
CH 3.3V
D_CSBD
SCLK
SDIO
SDO
AD9747BCPZRL
P1D5
P1D4
P1D3
P1D2
P1D1
P1D0
DCO
DNC
D3.3V
DVSS
QSEL
DNC
P2D15
P2D14
P2D13
P2D12
P2D11
P2D10
D-DVDD3.3V
RF5
R33
0Ω
A+4.5V
C14 100nF
VOUT
C1 10uF
R26
0Ω
R9
10K
DAC3
D-AVDD3.3V
上拉I2C
U7
VDD
VOUT
NC
AGND
SPI2C
VREFIO
NC
SDIN/SDA
SYNC#/A0
SCLK/SCL
DAC60501MDGSR
C15 220nF
SDIN/SDA
SCLK/SCL
R10
4.7K
R11
4.7K
D-AVDD3.3V
SDIN/SDA
SCLK/SCL

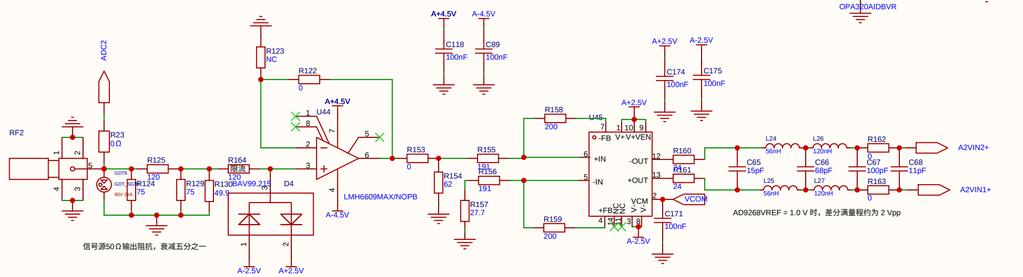

A
B
U33
SN74AVC16T245DGGR
R83
4.7K
DIR
A-DVDD1.8V
ADC_A_D0(LSB)
ADC_A_D1
ADC_A_D2
ADC_A_D3
ADC_A_D4
ADC_A_D5
ADC_A_D6
ADC_A_D7
ADC_A_D8
ADC_A_D9
ADC_A_D10
ADC_A_D11
ADC_A_D12
ADC_A_D13
ADC_A_D14
ADC_A_D15
CH-3.3V
C125
C124
100nF
C121
C120
ADC_DATA_OE_N
R87
D0A(LSB)
D1A
D2A
D3A
D4A
D5A
D6A
D7A
D8A
D9A
D10A
D11A
D12A
D13A
D14A
D15A
U36
SN74AVC16T245DGGR
D0B(LSB)
D1B
D2B
D3B
D4B
D5B
D6B
D7B
D8B
D9B
D10B
D11B
D12B
D13B
D14B
D15B
C127
C126
C123
C122
U39
SN74AVC4T774PWR
C128
C129
ADC_ORA
ADC_OCOA
ADC_OCOB
ADC_ORB
ADC_CTRL_OE_N
R27
ORA
OCOA
OCOB
ORB



输入范围±20V

±3.22V

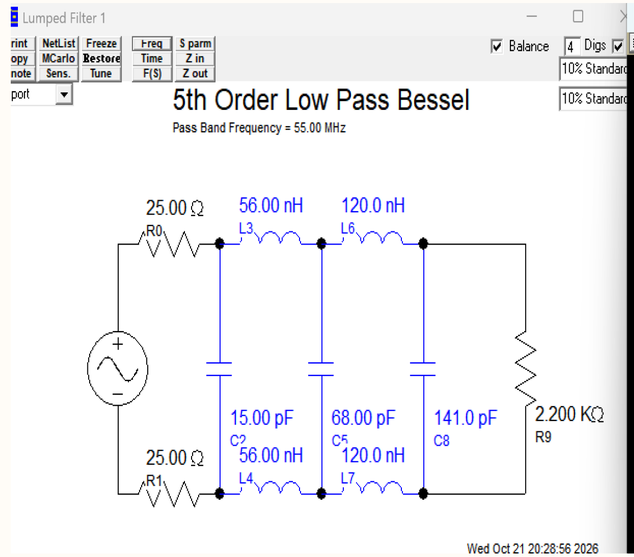
10VPP 50 欧 2VPP 2VPP

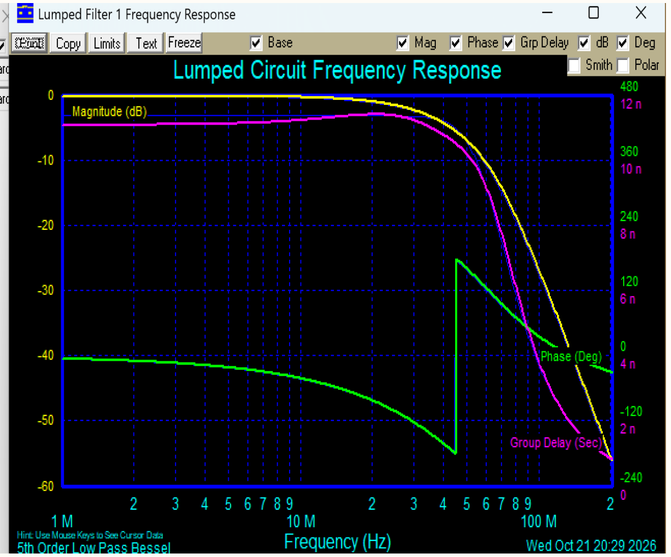


输入范围±20V

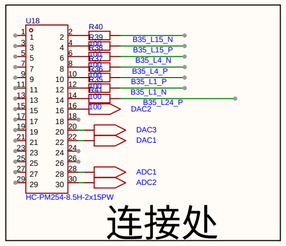
±3.22V

10VPP 50 欧 2VPP 2VPP

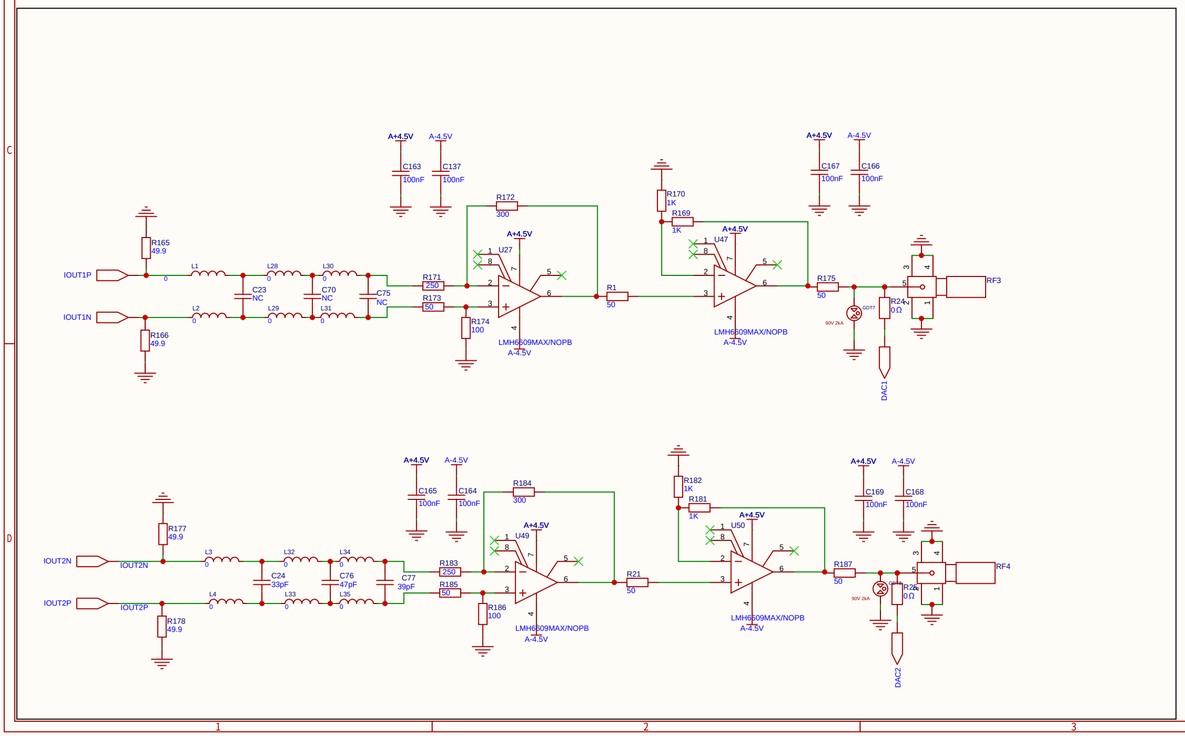


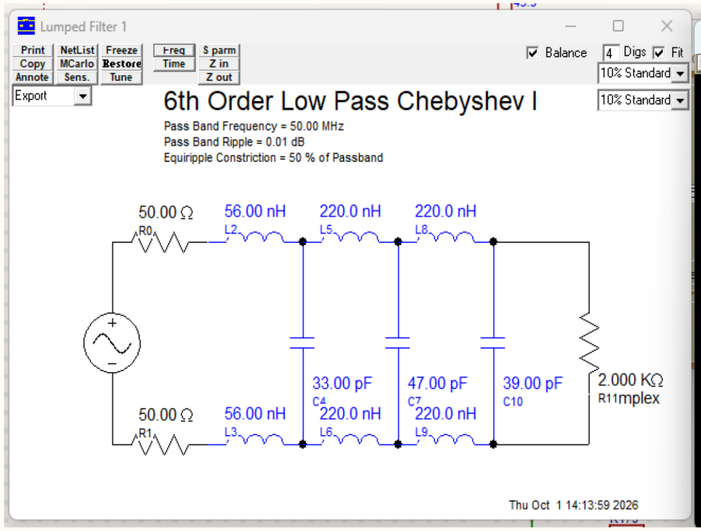


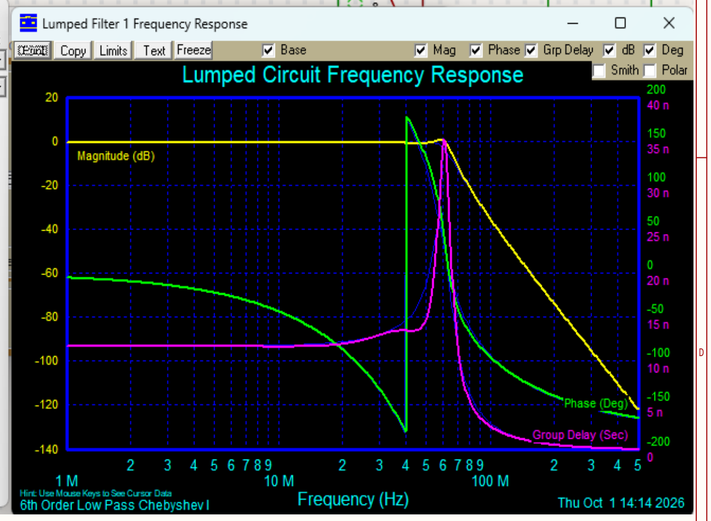
20M平坦



连接处







20 V PD输入
- DCDC +5.0 V
  - Zynq核心板
    - 5.0 W / 1.00 A
  - TPS7A2045 → +A4.5 V → 6×LMH6609正电源
    - 51 mA；负载0.230 W；+5V侧0.255 W
  - TPS7A2033 → A3.3 V → AD9747 AVDD33
    - 60 mA；负载0.198 W；+5V侧0.300 W
  - TPS7A2033 → D3.3 V → AD9747 DVDD33
    - 16 mA；负载0.053 W；+5V侧0.080 W
  - TPS7A2033 → CH-3.3 V
    - CH340
    - 电平转换器B侧
      - 按200 mA预算；负载0.660 W；+5V侧1.000 W
  - TPS7A4701 → DCLK3.3 V → AD9516-3
    - 典型：约0.36 A；负载约1.2 W；+5V侧约1.82 W
    - 最坏：约0.70 A；负载约2.3 W；+5V侧约3.49 W

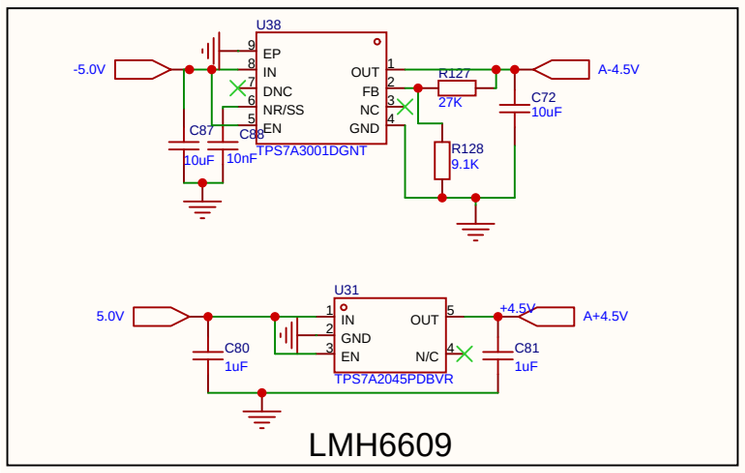
DCDC +5V合计：
- 约8.46～10.13 W，约1.69～2.03 A

- DCDC +3.0 V
  - TPS7A2025 → +A2.5 V → 2×LMH6554正电源
    - 114 mA；负载0.285 W；+3V侧0.342 W
  - TPS7A8801 OUT1 → A-AVDD1.8 V → AD9268 AVDD
    - 400 mA；负载0.720 W；+3V侧1.200 W
  - TPS7A8801 OUT2 → A-DVDD1.8 V
    - AD9268 DRVDD：55 mA
    - 3片电平转换器VCCA：按200 mA预留
      - 合计255 mA；负载0.459 W；+3V侧0.765 W
  - TPS7A2018 → AD9747 CVDD18
    - 22 mA；负载0.040 W；+3V侧0.066 W
  - TPS7A2018 → AD9747 DVDD18
    - 36 mA；负载0.065 W；+3V侧0.108 W

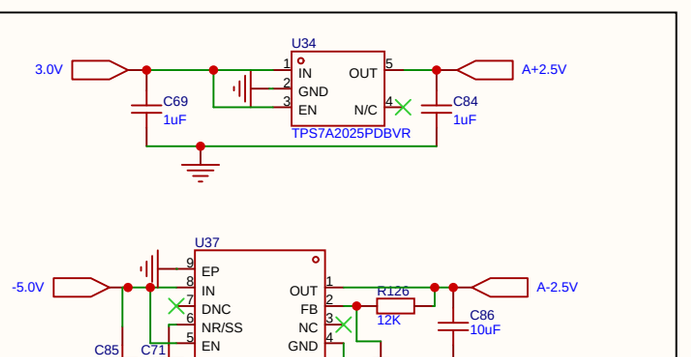
DCDC +3V合计：
- 约2.48 W，约0.83 A

- DCDC 5.0 V
  - TPS7A3001 → A4.5 V → 6×LMH6609负电源
    - 51 mA；负载0.230 W； 5V侧0.255 W
  - TPS7A3001 → A2.5 V → 2×LMH6554负电源
    - 114 mA；负载0.285 W； 5V侧0.570 W

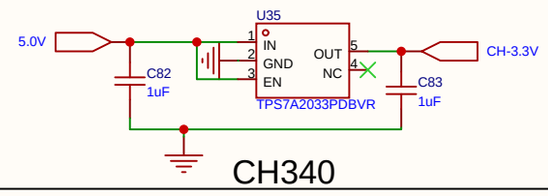
DCDC 5V合计：
- 约0.825 W，约0.165 A



LMH6609

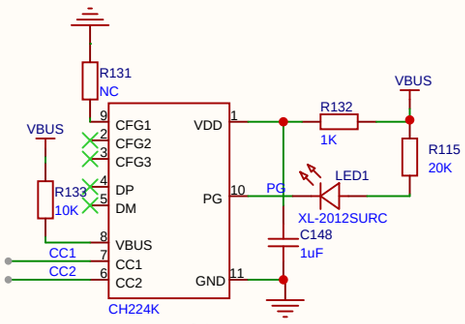


LMH6554

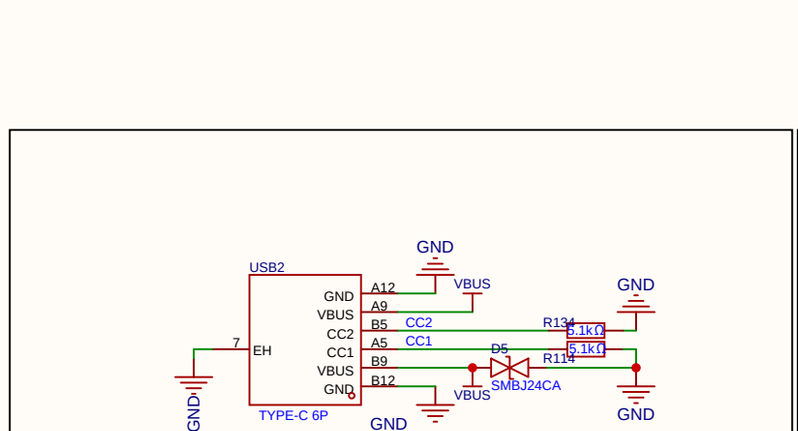


CH340

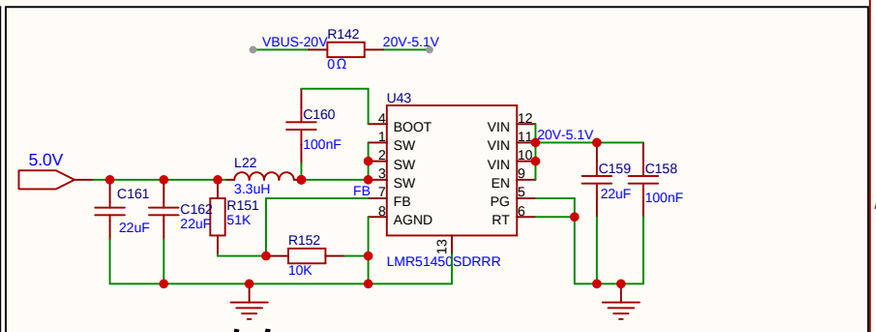
TYPE-C接口



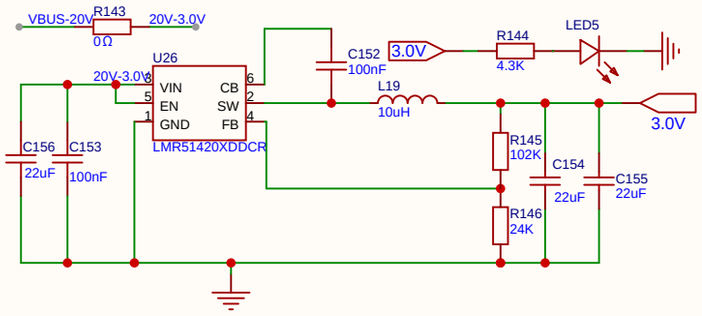
PD协议



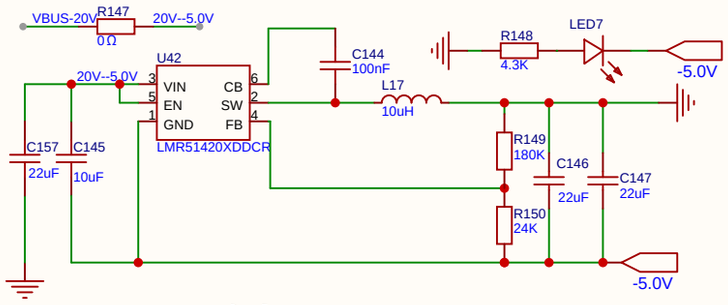
软启动、过压



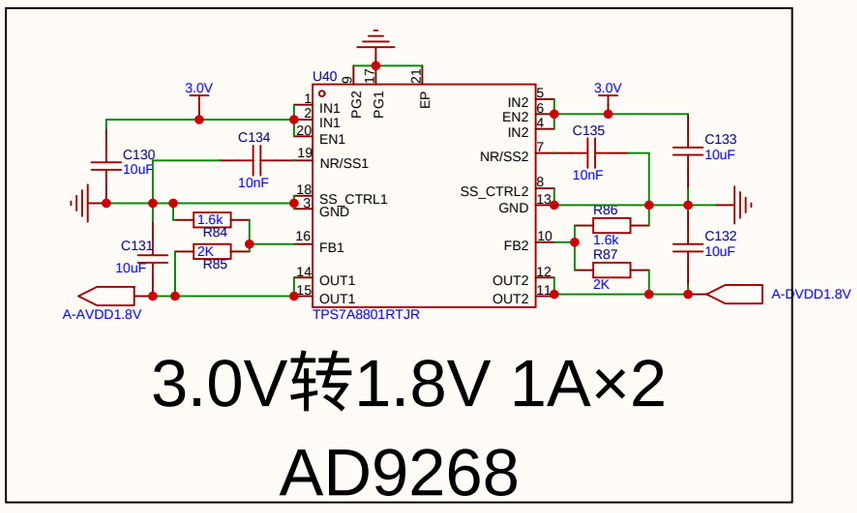
20V转5.0V 4A



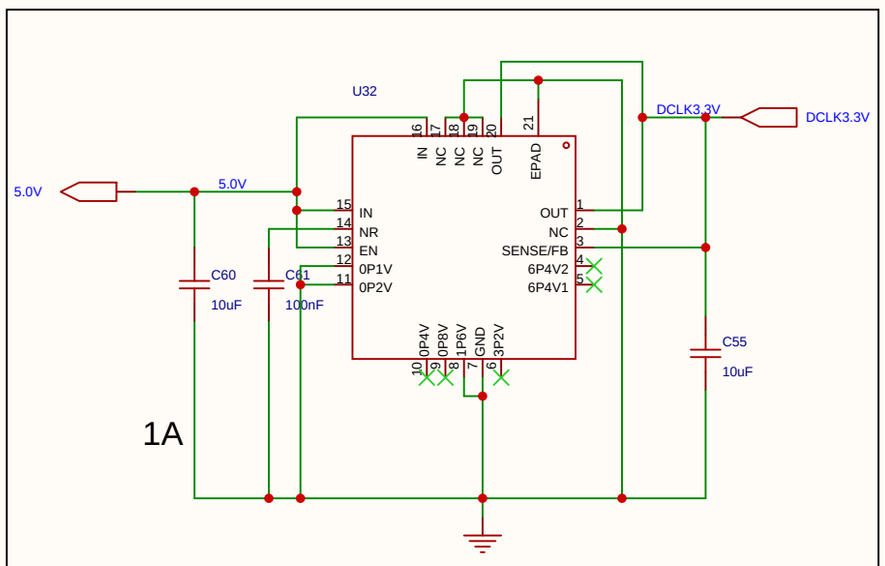
20V转3V 1.5A



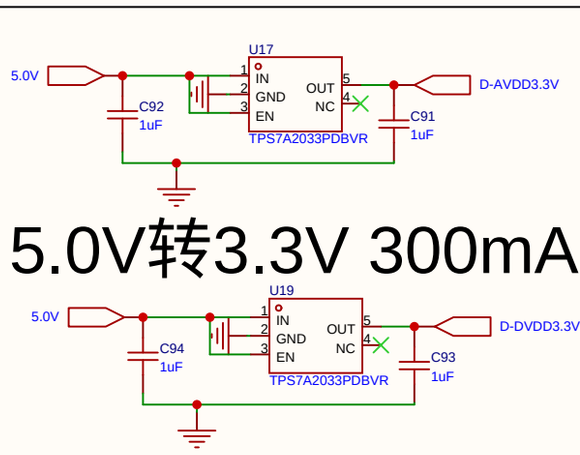
20V转-5V 1A



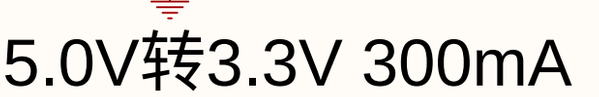
3.0V转1.8V 1A×2
AD9268



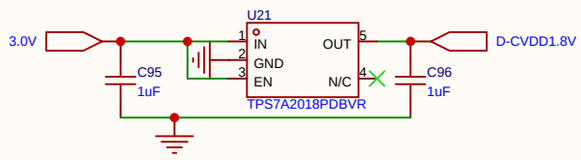
5.0V转3.3V 1A
AD9516-3



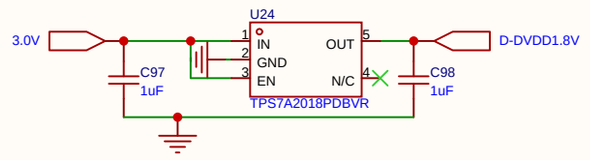
5.0V转3.3V 300mA



5.0V转3.3V 300mA



3.0V转1.8V 300mA



3.0V转1.8V 300mA
AD9747